These are algebraic ingredients for perturbation-indexed well groups. The ambient space and scalar structure are fixed explicitly. This file does not assert the two-extreme-perturbation characterization.

Result 1 (indexed intersection membership). [Mathlib, Submodule.mem_sInf]
Let K, V and I be types. Let f be a field structure on K. Let L be a vector space structure on V over f. Let A be a subset of I. Let S be a function from I to Subspace(L). Let x be an element of V. Then

$$x \text{ belongs to } \bigcap_{i \in A} S(i) \text{ if and only if } (\text{for every } i : I,\ i \text{ belongs to } A \text{ implies } x \text{ belongs to } S(i)).$$

Adapted from the cited source.

Result 2 (intersection is included in each member). [Mathlib, Submodule.mem_sInf]
Let K, V and I be types. Let f be a field structure on K. Let L be a vector space structure on V over f. Let A be a subset of I. Let S be a function from I to Subspace(L). Let i be an element of I. Assume i belongs to A (`admissible index`). Then

$$\bigcap_{i \in A} S(i) \text{ is a subspace of } S(i).$$

Adapted from the cited source.

Result 3 (more indices give a smaller intersection). [Mathlib, Submodule.mem_sInf]
Let K, V and I be types. Let f be a field structure on K. Let L be a vector space structure on V over f. Let A be a subset of I. Let S be a function from I to Subspace(L). Let B be a subset of I. Assume A is a subset of B (`more indices`). Then

$$\bigcap_{i \in B} S(i) \text{ is a subspace of } \bigcap_{i \in A} S(i).$$

Adapted from the cited source.

Result 4 (empty family contains every vector). [Mathlib, Submodule.mem_sInf]
Let K, V and I be types. Let f be a field structure on K. Let L be a vector space structure on V over f. Let A be a subset of I. Let S be a function from I to Subspace(L). Let x be an element of V. Assume (for every $i : I$, not (i belongs to A)) (`no admissible indices`). Then

$$x \text{ belongs to } \bigcap_{i \in A} S(i).$$

Adapted from the cited source.

LLM-assisted Clean specimens and proofs (gpt-6, 2026-11-06). Mathematical sources are attributed above; see PROVENANCE.md.